

RAKSHITA N

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Professional Summary

Software Systems student skilled in Python and C++ with a strong foundation in Data Structures, Algorithms, Data Analysis, and AI. Experienced in working with data-driven projects, architecting robust data pipelines, and deploying Machine Learning and Generative AI solutions. A quick learner with a deep interest in predictive modeling, AI innovation, data visualization, and extracting meaningful, actionable insights from complex real-world data.

Education

M. Sc. Software Systems (5 years integrated), <i>Coimbatore Institute of Technology</i>	2024 – 2029
CGPA: 8.3 [Till 4th sem]	
Class X, <i>Sri Chaitanya Techno School</i>	2022 – 93%
Class XII, <i>Sri Chaitanya Techno School</i>	2024 – 81%

Projects

AgriHive AI | NIT Trichy Hackathon

- Built a privacy-preserving agricultural intelligence platform enabling collaborative learning across farms using Feature-Aware Personalized Federated Learning (FAP-FL).
- Designed a modular AI pipeline integrating Random Forest, Federated Learning, Digital Twin Simulation, PSO Optimization, and Explainable AI (SHAP) for personalized agricultural risk assessment.
- Developed a scalable full-stack solution with Python, FastAPI, React, PostgreSQL, SQLAlchemy, and external weather APIs.

MOTION-AI | Python, XGBoost, Keras/LSTM, PyTorch, SHAP, FastAPI

- Predictive Maintenance System: Developed XGBoost and LSTM (Keras) models for an industrial AI cockpit to predict immediate machine failures and forecast Remaining Useful Life (RUL).
- Implemented SHAP to provide transparent, interpretable model outputs, allowing operators to understand key drivers of predicted machine failures.
- Integrated computer vision pipelines using PyTorch to perform heuristic visual analysis on cutting tools and component wear.
- Deployed ML models via a FastAPI backend for low-latency, real-time sensor inference.

University Knowledge Base | Python, LangChain, RAG, ChromaDB, Streamlit

- Developed an AI-powered University Knowledge Base using **Python, LangChain, RAG, ChromaDB, and Streamlit**, enabling context-aware question answering from university documents.
- Implemented document ingestion, text chunking, local embeddings, vector search, and OCR support for efficient knowledge retrieval.

SkyPulse | Python, Snowflake, Apache Kafka, Airflow, dbt, Pandas, Docker

- Architected a real-time hybrid batch/streaming pipeline to ingest global flight telemetry into Snowflake, serving as the foundational data layer for analytics and ML use cases.
- Built a Medallion data architecture with automated data validation and feature extraction, ensuring high-quality inputs for downstream modeling.
- Developed a comprehensive data quality auditing framework and surfaced real-time metrics via a custom FastAPI and React dashboard.

Technical Skills

Programming Languages

C, C++, Python, Java

Web Technologies

HTML, CSS, JavaScript, Bootstrap

Frameworks & Tools

FastAPI, Next.js, React, Git, GitHub

Machine Learning

Scikit-Learn, XGBoost, PyTorch, Pandas, NumPy

Databases

MySQL, SQLite, Vector Databases (Chroma), MongoDB, dbt

Artificial Intelligence

RAG, LangChain, Generative AI

Certificates

- Revature FullStack Development Certificate
- Revature Generative AI training program

Achievements

1st place Tata Centre for Artificial Intelligence and Machine Learning NIT Trichy

2nd Prize – Revature Capstone Project

Certificate of Excellence – Best Theme, *Rotaract Club of Coimbatore Uptown*
Rac-a-Thon Hackathon